

Hybrid Neural Network Based Signal Detection for Multi-User MIMO with Hardware Imperfections

Dawei Gao, Wenfeng Zhang, Qinghua Guo, Guisheng Liao, and Ming Jin

Abstract—This work deals with the issue of communication signal detection in multi-user multi-input multi-output (MIMO) with hardware imperfections, where low-cost users suffer from power amplifier nonlinearity (PAN) and in-phase and quadrature imbalance (IQI). To deal with the multi-user interference and nonlinear distortions due to hardware imperfections, we propose a novel signal detector using a hybrid neural network (HNN), which combines a Kolmogorov-Arnold network (KAN) and a conventional linear NN layer. The training of the HNN involves the update of both linear NN weights and nonlinear activation functions. It turns out that the proposed HNN detector with a much smaller model size significantly outperforms conventional polynomial-based techniques and state-of-the-art pure NN or KAN techniques.

Index Terms—Hardware imperfections, Kolmogorov-Arnold networks, multiple-input multiple-output, nonlinear distortion.

I. INTRODUCTION

AS a core technology of modern wireless communication systems, multiple-input multiple-output (MIMO) significantly enhances channel capacity and transmission reliability by effectively utilizing the spatial dimension. However, as communication systems evolve toward higher frequency bands, larger bandwidths, and more complex modulation schemes, hardware impairments in low-cost radio frequency (RF) front-ends have become increasingly prominent, posing a critical bottleneck to further performance improvement in MIMO systems [1], [2], [3]. Hardware impairments in low-cost RF front-ends significantly degrade the performance of MIMO systems, especially in higher frequency bands and complex modulation schemes [4].

Among various hardware impairments, in-phase/quadrature (I/Q) imbalance is one of the most common types, manifesting as mismatches between the I and Q signal paths, thereby disrupting the orthogonality of the transmitted signals [5]. Meanwhile, the nonlinear characteristics of power amplifiers (PAs) leading to severe signal distortion cannot be overlooked, which is particularly pronounced in millimeter-wave and terahertz communication systems [3]. The combined effects of these hardware impairments further exacerbate signal distortion, posing significant challenges to the receiver design of MIMO systems. To address the distortion issues caused by

hardware impairments, various solutions have been proposed. For the nonlinear distortion caused by the PA, some classic polynomials-based methods have been investigated [6], [7]. These methods approximate the nonlinear characteristics of power amplifiers (PAs) by constructing polynomial models, and apply either predistortion at the transmitter to compensate for the nonlinear effects introduced by PAs, or postdistortion at the receiver to mitigate the distortion [8]. However, these approaches are prone to the numerical instability issue due to the matrix inversion in the determination of polynomial coefficients [9]. Other methods using the Wiener model [10] and the Hammerstein model [11] have also been studied, however, these approaches often result in model mismatch issues. Moreover, most existing works address I/Q imbalance (IQI) or PA nonlinearity (PAN) individually [12], [13], despite their concurrent occurrence in many cases, which necessitates handling their combined effects.

Methods based on artificial neural networks (NNs) have demonstrated significant potential in addressing nonlinear distortion in communication systems, thanks to their universal approximation capabilities. In [14], a parallel multi-task learning (MTL) estimator is developed to mitigate the coupling effects among different hardware impairments by utilizing their mutual correlations, which proves effective in compensating IQI. The work in [15] first introduced a real-valued focused time-delay neural network (RVFTDNN) to simulate PA's behavior, achieving promising performance. Building on this, extensive research [16], [17] has explored enhanced versions of RVFTDNN. Among these, the augmented RVFTDNN (ARVTDNN) model [16] improves the mitigation of PAN and IQI by incorporating higher-order signal components into the input of RVFTDNN. However, an issue with the ARVTDNN based techniques is that a large number of pilot symbols are required to train it properly, leading to unacceptable overhead and hindering their applications especially in time varying environments.

Most recently, Kolmogorov-Arnold networks (KANs) have been proposed [18], based on the Kolmogorov-Arnold representation theorem, which decomposes high-dimensional functions into low-dimensional combinations through a unique structure, achieving efficient nonlinear function approximation. It motivates this work, aiming to build a more efficient network with smaller size for signal detection, enabling the training of the network with less pilot symbols but detection with higher performance. We will show that the use of a pure KAN is not efficient in dealing with the multi-user interference (MUI) in the MIMO system. Considering that the MUI is linear, we incorporate a linear NN layer to the KAN, resulting in a hybrid NN-based signal detector, which handles the combined effects of MUI, IQI and PAN. The linear layer

Corresponding Author: Qinghua Guo (qguo@uow.edu.au).

Dawei Gao, Wenfeng Zhang and Guisheng Liao are with the Hangzhou Institute of Technology, Xidian University, Hangzhou 311200, China (e-mail: gaodawei@xidian.edu.cn; 24241214808@stu.xidian.edu.cn; liaogs@xidian.edu.cn).

Qinghua Guo is with the School of Electrical, Computer and Telecommunications Engineering, University of Wollongong, NSW 2522, Australia (e-mail: qguo@uow.edu.au).

Ming Jin is with the Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo 315211, China (email: jinming@nbu.edu.cn).

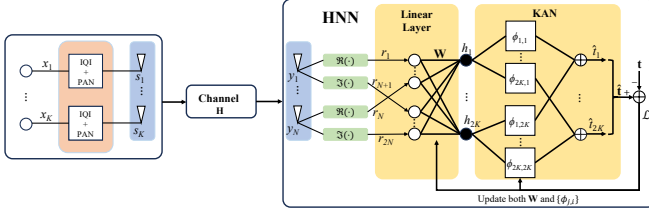


Fig. 1: Block diagram of the multi-user MIMO system with PAN and IQI, where the HNN is proposed for signal detection.

also significantly reduces the dimension of the signal, which is input to the KAN, leading to more efficient NN architecture. At the network training level, it is worth noting that traditional neural networks learn node weights, whereas KANs focus on learning activation functions [18]. The proposed HNN, which combines the linear layer of traditional NNs with KAN, can be trained within the backpropagation framework to achieve joint learning of both activation functions and linear weights. The derivations of the gradient backpropagation algorithm for the proposed HNN is presented in Section VI. Simulation results demonstrate that our HNN-based detector significantly outperforms existing detectors.

II. SYSTEM MODEL

Consider uplink transmission in a multi-user millimeter-wave MIMO system with K single-antenna user devices, each equipped with low cost modulators and PAs, that introduce PAN and IQI. At the receiver end (e.g., a base station (BS) or access point (AP)), N antennas are deployed

Define vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_K(t)]^T$ as the signal symbols transmitted by K users at a given instant t , where $x_k(t)$ denotes the symbol transmitted by the k -th user at time t . The signal due to the impact of IQI can be represented as [17]

$$x_k^I(t) = \alpha_k x_k(t) + \beta_k x_k^*(t), \quad (1)$$

where

$$\alpha_k = \cos(a_k/2) + j b_k \sin(a_k/2) \quad (2)$$

$$\beta_k = b_k \cos(a_k/2) + j \sin(a_k/2), \quad (3)$$

a_k is the phase imbalance parameter, and b_k is the amplitude imbalance parameter. The imbalanced signal is then fed into the PA.

The nonlinear distortion of the power amplifier consists of two parts, i.e., $\psi(|x_k^I(t)|)$ and $\eta(|x_k^I(t)|)$ [19]. Among them, $\psi(|x_k^I(t)|)$ describes the amplitude-to-amplitude (AM/AM) conversion, i.e.,

$$\psi(|x_k^I(t)|) = \frac{\alpha_\psi |x_k^I(t)|}{(1 + (\alpha_\psi \frac{|x_k^I(t)|}{\lambda_\psi})^{2\sigma_\psi})^{\frac{1}{2\sigma_\psi}}}, \quad (4)$$

while $\eta(|x_k^I(t)|)$ describes the amplitude-to-phase (AM/PM) conversion, i.e.,

$$\eta(|x_k^I(t)|) = \frac{\alpha_\eta |x_k^I(t)|^{m_1}}{1 + (\frac{|x_k^I(t)|}{\lambda_\eta})^{m_2}}, \quad (5)$$

where α_ψ denotes the linear gain, λ_ψ denotes the normalized saturation output amplitude, σ_ψ and α_η denote smoothness factor and phase distortion scaling factor, respectively, λ_η denotes phase distortion threshold, and m_1 and m_2 are fitting

orders of the nonlinear model. The distorted signal can then be expressed as

$$s_k(t) = \psi(|x_k^I(t)|) e^{j(\text{angle}(x_k^I(t)) + \eta(|x_k^I(t)|))}, \quad (6)$$

where $\text{angle}(x_k^I)$ denotes the phase of the complex signal x_k^I . The received signal at time instant t is represented as

$$\mathbf{y}(t) = \mathbf{H}\mathbf{s}(t) + \mathbf{n}(t), \quad (7)$$

where $\mathbf{H} \in \mathbb{C}^{N \times K}$ is the MIMO channel matrix, $\mathbf{y}(t) = [y_1(t), y_2(t), \dots, y_N(t)]^T$, $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$ and $\mathbf{n}(t)$ denotes a white Gaussian noise vector.

Since the signals used in the above model are complex-valued, we separate the real and imaginary parts of the signal and rearrange them into a real-valued form, i.e.,

$$\begin{bmatrix} \Re\{\mathbf{y}(t)\} \\ \Im\{\mathbf{y}(t)\} \end{bmatrix} = \underbrace{\begin{bmatrix} \Re\{\mathbf{H}\} & -\Im\{\mathbf{H}\} \\ \Im\{\mathbf{H}\} & \Re\{\mathbf{H}\} \end{bmatrix}}_{\mathbf{H}'} \underbrace{\begin{bmatrix} \Re\{\mathbf{s}(t)\} \\ \Im\{\mathbf{s}(t)\} \end{bmatrix}}_{\mathbf{s}'(t)} + \underbrace{\begin{bmatrix} \Re\{\mathbf{n}(t)\} \\ \Im\{\mathbf{n}(t)\} \end{bmatrix}}_{\mathbf{n}'(t)}. \quad (8)$$

The input-output relationship of the MIMO system is complex due to the combined impacts of IQI, PAN and MUI which can be expressed as

$$\mathbf{r}(t) = \mathcal{S}(\mathbf{x}(t)) + \mathbf{n}'(t), \quad (9)$$

where $\mathcal{S}(\cdot)$ represents the system transfer function. The channel matrix, as well as the parameters of the IQI and PAN models, are assumed to be unknown. Each user transmits a pilot signal followed by data. The primary objective of the receiver at the BS/AP is to accurately detect the transmitted data symbols from all users.

III. PROPOSED HNN RECEIVER

We propose a HNN receiver based on KAN as shown in Fig. 1. Instead of using a pure KAN, we incorporate a conventional NN linear layer into KAN, which is more effective to handle the linear MUI introduced by the MIMO channel matrix. The linear layer also significantly reduces the dimension of the signals as explained later. The linear layer is then followed by a KAN to handle the nonlinear distortions due to IQI and PAN. This leads to the HNN structure, which is used to learn the inverse function of $\mathcal{S}(\cdot)$ in (9) and jointly mitigate the MUI and nonlinear distortions. It is noted that the linear layer learns weight parameters directly, whereas the KAN module learns nonlinear activation functions. Despite this distinction, the overall HNN architecture remains fully trainable within the standard backpropagation framework.

The N -dimensional received signal is first processed by real and imaginary separation before being input into the HNN model. As shown in Fig. 1, the linear layer is used to deal with the linear MUI of the channel, yielding

$$\mathbf{h} = \mathbf{W}\mathbf{r}, \quad (10)$$

where $\mathbf{h} = [h_1, h_2, \dots, h_{2K}]^T$, $\mathbf{r} = [r_1, r_2, \dots, r_{2N}]^T$, and the weight matrix $\mathbf{W}$ has the size $2K \times 2N$. Considering the structure of $\mathbf{H}'$ in (8), we tie the entries of the weight matrix $\mathbf{W}$, so that we have

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}^1 & \mathbf{W}^2 \\ -\mathbf{W}^2 & \mathbf{W}^1 \end{bmatrix}, \quad (11)$$

where $\mathbf{W}^1$ and $\mathbf{W}^2$ are matrices with dimension $K \times N$. The output of the linear layer contains $2K$ neurons, which

also serve as the input of the subsequent KAN layer. Since K is typically much smaller than N in the system model, the linear layer also plays a role in data dimensionality reduction, reducing the dimensionality of the data input to the KAN layer. This significantly decreases the total number of model parameters, which will be demonstrated later.

After the signal is output from the linear layer, we assume that it has eliminated linear interference and contains only nonlinear distortions. Then, the distorted signal is input to the KAN layer, and the output from the j th node of KAN layer is represented as

$$\hat{t}_j = \sum_{i=1}^{2K} \phi_{j,i}(h_i), j = 1, 2, \dots, 2K, \quad (12)$$

where $\phi_{j,i}(\cdot)$ is the activation function connecting node h_i and $\hat{t}_j$. It is composed of a spline function and a basis function component $b(h_i)$ that serves a role similar to a residual connection [18]. Specifically, the activation function is defined as

$$\phi_{j,i}(h_i) = \omega_s^{j,i} \text{spline}^{j,i}(h_i) + \omega_b^{j,i} b(h_i), \quad (13)$$

where the basis function $b(h_i)$ typically adopts the sigmoid linear unit (SiLU), defined as

$$\text{SiLU}(h_i) = \frac{h_i}{1 + e^{-h_i}}, \quad (14)$$

and the spline function $\text{spline}^{j,i}(h_i)$ is a weighted sum of multiple nonlinear k -th order B-spline basis functions $B_z(h_i)$ [18]:

$$\text{spline}^{j,i}(h_i) = \sum_{z=1}^Z c_z^{j,i} B_z(h_i), \quad (15)$$

where $Z = \text{grid} + k$, both grid and k are model parameters, and the weights ω_s , ω_b and c_z are trainable parameters. Hence, the output signal of KAN layer can be written in the following form

$$\hat{\mathbf{t}} = \Phi(\mathbf{h}) = \left[\sum_{i=1}^{2K} \phi_{1,i}(h_i), \dots, \sum_{i=1}^{2K} \phi_{2K,i}(h_i) \right]^T, \quad (16)$$

where $\hat{\mathbf{t}} = [\hat{t}_1, \hat{t}_2, \dots, \hat{t}_{2K}]^T$. Finally, by combining the estimated real and imaginary parts, the estimate of the transmitted signal can be reconstructed as $\hat{x}_k = \hat{t}_{2k-1} + j\hat{t}_{2k}$, $k = 1, 2, \dots, K$. The above design enables the HNN to transform the problem of fitting complex multivariate nonlinear functions into fitting multiple low-dimensional univariate functions. Through the superposition and combination of univariate functions, the proposed HNN can represent complex nonlinear functional relationships with higher precision, making it suitable for addressing combined effects of linear MUI due to the channel and nonlinear distortions induced by hardware impairments.

The training of the HNN can be implemented using back propagation. Suppose there are M_0 training samples $\{(\mathbf{r}(m), \mathbf{t}(m)), m = 1, 2, \dots, M_0\}$, where $\mathbf{t}(m) = [t_1(m), t_2(m), \dots, t_{2K}(m)]^T$ denotes the training symbols with $t_{2i-1} = \Re\{x_i\}$ and $t_{2i} = \Im\{x_i\}$, for $i = 1, 2, \dots, K$. The corresponding received signal is given by $\mathbf{r}(m) = [r_1(m), r_2(m), \dots, r_{2N}(m)]^T$. The loss function can be written as

$$\mathcal{L} = \frac{1}{M_0} \sum_{m=1}^{M_0} E(m), \quad (17)$$

where $E(m) = \frac{1}{2K} \sum_{i=1}^{2K} [t_i(m) - \hat{t}_i(m)]^2$. The loss function is calculated to update both the weight $\mathbf{W}$ and activation functions $\{\phi_{j,i}\}$ using back propagation.

IV. SIMULATION RESULTS

Assume that there are $K = 2$ users, each equipped with one antenna, and $N = 32$ antennas at the receiver end. Due to the limited space in this correspondence, we consider a flat-fading MIMO channel and note that the proposed method can be readily extended to frequency selective channels leveraging the orthogonal frequency division multiplexing technique. We use 16-quadrature amplitude modulation (QAM) in the simulations. For the IQI parameters, we set $a_k = 4^\circ$ and $b_k = 0.05$; for the PAN parameters, we set $\alpha_\psi = 4.65$, $\lambda_\psi = 0.58$, $\sigma_\psi = 0.81$, $\alpha_\eta = 2560$, $\lambda_\eta = 0.114$, $m_1 = 2.5$ and $m_2 = 2.3$ [19]. The experiments are implemented using the deep learning framework PyTorch. All simulations are conducted on a workstation equipped with the Microsoft Windows 11 operating system, an Intel Core i7-14700 processor, and an NVIDIA GeForce RTX 4060 GPU (8 GB). All NNs' training and inference are GPU-accelerated. A dataset consisting of 6000 samples is generated and it is split into training and testing sets in an 8:2 ratio. During training, we utilize the batch gradient descent strategy to optimize the model parameters with batch size 128. The adaptive moment estimation (Adam) optimizer is employed for parameter training with learning rate 0.0005. For the KAN layer of the proposed receiver, a single-hidden layer with 4 neurons is employed, with the model parameters setting $\text{grid} = 5$ and $k = 2$.

We evaluate the mean squared error (MSE) and symbol error rate (SER) performance of various methods under signal-to-noise ratio (SNR) conditions ranging from -8 to 8 dB. The evaluated methods include: (a) the proposed HNN; (b) pure KAN; (c) ARVTDNN [16]; (d) the minimum mean square error (MMSE)+KAN method (MMSE for channel equalization and KAN for dealing with nonlinearity); and (e) the real-valued polynomial (RP) method with a fifth order [20]. The left half of Fig. 2 illustrates the SER performance of these methods. It can be observed that the RP based method does not perform well due to the numerical instability. Even with a known channel matrix $\mathbf{H}$, the RP base method has poor SER performance compared to others. At the same time, the ARVTDNN has better performance, while it still has large gaps to the proposed HNN method. The MMSE+KAN method shows a modest performance improvement over pure KAN, suggesting that MMSE contributes to the mitigation of MUI. However, it still underperforms compared to the proposed HNN. This is because calculating MMSE weights requires a known nonlinearity, which is unavailable in practice, leading to a model mismatch issue that affects its performance. Meanwhile, the HNN method has over 2dB performance gain compared to the pure KAN, thanks to the introduction of the linear layer, which makes HNN more efficient in handling the combined effects of both linear MUI and nonlinear distortions. It is worth mentioning that the HNN method also has low complexity due to its small model size as shown in Table I. The right half of Fig. 2 shows the MSE performance of the

TABLE I: Computational Complexity and Latency Analysis of Various Receivers.

Network	Dimensions	No. of Para.	FLOPs (Training)	Latency(s)		Complexity	
				Training	Inference	Training	Inference
HNN	[64,4,4]	468	7.95×10^8	68.1909	0.2446	$\mathcal{O}(ES[N_{in}N_{mid} + N_{mid}N_{out}M])$	$\mathcal{O}(N_{in}N_{mid} + N_{mid}N_{out}M)$
MMSE+KAN	[64,4,4]	464	2.62×10^8	51.2003	0.2892	$\mathcal{O}(N_{in}^2S + N_{in}^3 + ESN_{mid}N_{out})$	$\mathcal{O}(N_{in}N_{mid} + N_{mid}N_{out}M)$
Pure KAN	[64,4,4]	3536	4.11×10^9	874.9	8.0640	$\mathcal{O}(ES[N_{in}N_{mid} + N_{mid}N_{out}]M)$	$\mathcal{O}(M[N_{in}N_{mid} + N_{mid}N_{out}])$
ARVTDNN	[480,15,4]	7279	1.57×10^{10}	5.3277	0.0631	$\mathcal{O}(ES[N_{in}N_{mid} + N_{mid}N_{out}])$	$\mathcal{O}(N_{in}N_{mid} + N_{mid}N_{out})$
RP	[576, 4]	2304	2.76×10^6	0.89	0.0206	$\mathcal{O}(SN_{in}^2)$	$\mathcal{O}(N_{in}N_{out})$

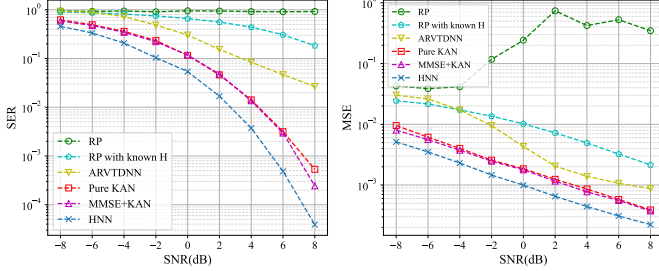
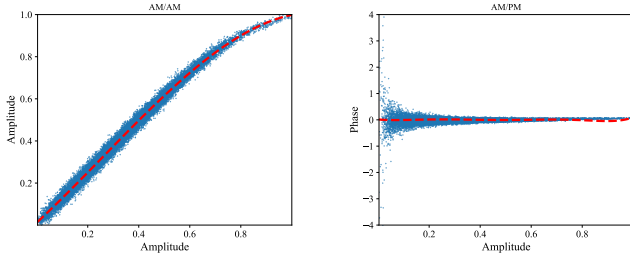


Fig. 2: SER and MSE performance comparisons of various receivers.

forementioned receivers. It can also be observed clearly that the proposed HNN method outperforms the state-of-the-art methods significantly.



(a) AM/AM distortion curve of the collected dataset. (b) AM/PM distortion curve of the collected dataset.

(c) SER performance comparisons of various receivers on the collected dataset.

Fig. 3: Distortion characteristics of the collected dataset and SER performance comparison.

To validate the performance of the proposed HNN on a more diverse dataset, we adopt the data collected in [21], which contains I/Q signals from commercial PAs. The distortion characteristics of this dataset are first fitted, and the fitted models are then used to generate new data for training and evaluating the SER performance of various receivers. Figs. 3a and 3b illustrates the AM/AM and AM/PM distortion characteristic curves fitted from this dataset. The results are presented in Fig. 3c. It can be observed that, in the presence of practical PA nonlinearity, the proposed HNN model still demonstrates superior performance. It validates that the proposed HNN

method is not confined to specific power amplifier nonlinearity models, but rather is a learning-based approach capable of adapting to the actual nonlinear characteristics of PAs.

Table I presents a comprehensive analysis of computational complexity, model size, and runtime latency. In the table, N_{in} , N_{mid} , and N_{out} representing the input, intermediate, and output dimensions of various receivers, respectively, which are detailed in the Dimension column of Table I. Additionally, M denotes the B-spline parameter for KAN layers, while E and S refer to the number of training epochs and the training sample size. It can be seen that the proposed HNN model demonstrates high training efficiency with only 468 trainable parameters. Despite sharing the same topology with the Pure KAN model, the HNN achieves a significant reduction compared to the 3,536 parameters required by the pure KAN baseline. This efficiency stems from replacing the parameter-intensive KAN layer with a standard linear layer. In contrast, the ARVTDNN baseline requires an expanded input dimension of 480 and a hidden dimension of 15 to capture time-domain features. Consequently, ARVTDNN necessitates 7,279 parameters, which is over 15 times that of the HNN. This comparison effectively demonstrates the structural compactness of the proposed model. Regarding computational speed, the RP method exhibits the lowest theoretical training complexity of $\mathcal{O}(SN_{in}^2)$ as it avoids backpropagation, whereas the ARVTDNN incurs the highest theoretical cost due to high-dimensional feature extraction. However, empirical results indicate that ARVTDNN currently achieves the lowest runtime latency among all NN-based models. This discrepancy is primarily attributed to the mature hardware acceleration available for traditional MLPs. KAN-based architectures, being in an early research stage, currently lack such specialized operator optimization for GPU. Nevertheless, the HNN model provides outstanding performance gain compared to other state-of-the-art methods while reduced complexity significantly.

V. CONCLUSION

We have proposed a novel hybrid NN model, which combines a linear NN layer and a KAN layer, to handle the combined effects of the linear MUI and nonlinear hardware distortions in multi-user MIMO systems. We have shown that, compared to state-of-the-art methods, the proposed HNN model can have a much smaller size, while with significantly improved performance. Simulation results demonstrate the merits of the proposed HNN receiver.

VI. APPENDIX

Backpropagation is employed to train the proposed HNN, where gradient descent is executed not only on the linear

weights at the edges, but also the trainable activation functions. Here we provide detailed derivations of the gradients in the hybrid network.

According to the chain rule, the partial derivative of the MSE E in (17) (we omit the time constant for simplicity) with respect to the nonlinear activation function $\phi_{j,i}(h_u)$ and its input h_u can be expressed as

$$\frac{\partial E}{\partial \phi_{j,i}(h_u)} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \phi_{j,i}(h_u)}, \quad (18)$$

where $\frac{\partial E}{\partial \hat{t}_v} = 1/K$ and $\phi_{j,i}(h_u)$ is composed of trainable coefficients $\omega_s^{v,u}$, $\omega_b^{v,u}$ and $c_n^{v,u}$. In order to calculate the derivative to the nonlinear activation, we need to decompose the calculations with respect to these coefficients. Then, the partial derivative of E with respect to $\omega_s^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial \omega_s^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \omega_s^{v,u}}, \quad (19)$$

where

$$\partial E / \partial \hat{t}_v = 1/K, \partial \hat{t}_v / \partial \omega_s^{v,u} = \text{spline}^{v,u}(h_u). \quad (20)$$

By combining equations (19) and (20), we obtain

$$\partial E / \partial \omega_s^{v,u} = 1/K \cdot \text{spline}^{v,u}(h_u). \quad (21)$$

Similarly, we can obtain the partial derivative of E with respect to $\omega_b^{v,u}$,

$$\partial E / \partial \omega_b^{v,u} = 1/K \cdot \text{b}(h_u). \quad (22)$$

The partial derivative of E with respect to $c_n^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial c_n^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}}, \quad (23)$$

where

$$\frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} = \omega_s^{v,u}, \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}} = B_n(h_u). \quad (24)$$

Thus, we have

$$\partial E / \partial c_n^{v,u} = 1/K \cdot \omega_s^{v,u} B_n(h_u). \quad (25)$$

Then, we can obtain partial derivative of E with respect to the element $w_{u,v}$ in the u -th row and v -th column of $\mathbf{W}$ can be expressed as

$$\frac{\partial E}{\partial w_{u,v}} = \frac{1}{K} \sum_{p=1}^{2K} \frac{\partial \hat{t}_p}{\partial h_u} \frac{\partial h_u}{\partial w_{u,v}}, \quad (26)$$

where

$$\frac{\partial \hat{t}_p}{\partial h_u} = \omega_s^{p,u} \frac{\partial \text{spline}^{p,u}(h_u)}{\partial h_u} + \omega_b^{p,u} \frac{\partial \text{b}(h_u)}{\partial h_u}, \frac{\partial h_u}{\partial w_{u,v}} = r_v \quad (27)$$

and

$$\partial \text{spline}^{p,u}(h_u) / \partial h_u = \sum_q c_q^{p,u} \partial B_q(h_u) / \partial h_u. \quad (28)$$

By combining equations (26), (27), and (28), we obtain

$$\begin{aligned} \frac{\partial E}{\partial w_{u,v}} = & \frac{1}{K} \sum_{p=1}^{2K} \left\{ \omega_s^{p,u} \sum_q \left[c_q^{p,u} \frac{\partial B_q(h_u)}{\partial h_u} \right] r_v \right. \\ & \left. + \omega_b^{p,u} \frac{\partial \text{b}(h_u)}{\partial h_u} r_v \right\} \end{aligned} \quad (29)$$

where $\text{b}(h_u)$ represents the SiLU function, and $B_q(h_u)$ denotes the B-spline function, which are both differentiable.

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